

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
import numpy as np

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import SimpleRNN, Dense

from tensorflow.keras.datasets import mnist

from tensorflow.keras.utils import to_categorical


# Load the MNIST dataset

(x_train, y_train), (x_test, y_test) = mnist.load_data()


# Normalize the pixel values

x_train = x_train.astype('float32') / 255.

x_test = x_test.astype('float32') / 255.


# Reshape the data to fit the RNN (batch, timesteps, features)

# In this example, we treat each row of the 28x28 image as a time step

x_train = x_train.reshape((x_train.shape[0], 28, 28))

x_test = x_test.reshape((x_test.shape[0], 28, 28))


# One-hot encode the labels

y_train = to_categorical(y_train, 10)

y_test = to_categorical(y_test, 10)


# Build the RNN model

model = Sequential()

model.add(SimpleRNN(50, input_shape=(28, 28), activation='relu'))

model.add(Dense(10, activation='softmax'))
```

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Compile the model

```
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
```

Train the model

```
model.fit(x_train, y_train, epochs=10, batch_size=128, validation_data=(x_test, y_test))
```

Evaluate the model

```
test_loss, test_acc = model.evaluate(x_test, y_test, verbose=2)
```

```
print(f'Test accuracy: {test_acc}')
```

Epoch 1/10

C:\Users\valer\anaconda3\Lib\site-packages\keras\src\layers\rnn\rnn.py:204: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(**kwargs)
```

469/469 ————— 5s 8ms/step - accuracy: 0.4673 - loss: 1.4604 - val_accuracy: 0.8585 - val_loss: 0.4468

Epoch 2/10

469/469 ————— 3s 7ms/step - accuracy: 0.8649 - loss: 0.4348 - val_accuracy: 0.9071 - val_loss: 0.3039

Epoch 3/10

469/469 ————— 3s 7ms/step - accuracy: 0.9067 - loss: 0.3180 - val_accuracy: 0.9206 - val_loss: 0.2805

Epoch 4/10

469/469 ————— 3s 7ms/step - accuracy: 0.9194 - loss: 0.2716 - val_accuracy: 0.9306 - val_loss: 0.2387

Epoch 5/10

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469/469 ————— 3s 7ms/step - accuracy: 0.9310 - loss:
0.2316 - val_accuracy: 0.9409 - val_loss: 0.2030

Epoch 6/10

469/469 ————— 3s 7ms/step - accuracy: 0.9379 - loss:
0.2123 - val_accuracy: 0.9380 - val_loss: 0.2108

Epoch 7/10

469/469 ————— 4s 8ms/step - accuracy: 0.9393 - loss:
0.2042 - val_accuracy: 0.9302 - val_loss: 0.2334

Epoch 8/10

469/469 ————— 4s 8ms/step - accuracy: 0.9427 - loss:
0.1936 - val_accuracy: 0.9449 - val_loss: 0.1802

Epoch 9/10

469/469 ————— 4s 8ms/step - accuracy: 0.9474 - loss:
0.1763 - val_accuracy: 0.9420 - val_loss: 0.1914

Epoch 10/10

469/469 ————— 4s 8ms/step - accuracy: 0.9457 - loss:
0.1817 - val_accuracy: 0.9425 - val_loss: 0.1943

313/313 - 1s - 3ms/step - accuracy: 0.9425 - loss: 0.1943

Test accuracy: 0.9424999952316284